

The Evolution of a Sign Language Recognition Pipeline: From LSTM to Spatial Realities

Introduction: The Vision and Pipeline

The objective of this project was to develop a robust, real-time Sign Language Recognition system. The core pipeline was designed to translate raw video feeds into classified text by tracking human movement. The system operated on a multi-stage data processing pipeline:

1. **Video Capture & Preprocessing:** Taking raw video, applying a strict 55% bounding-box crop to isolate the signer, and uniformly sampling exactly 30 frames per video.
2. **Feature Extraction:** Utilizing MediaPipe Holistic to map 225 specific 3D coordinates across the face, left hand, and right hand.
3. **Normalization:** Translating those coordinates to a central origin (the nose) and dividing by shoulder width to make the math theoretically immune to the user's distance from the camera.
4. **Sequence Prediction:** Feeding that 30-frame sequence into a Long Short-Term Memory (LSTM) neural network to predict the final sign.

Why This Approach Was Chosen

The combination of MediaPipe and LSTM was initially selected because it represents a highly logical baseline for temporal data classification. MediaPipe is exceptionally lightweight and provides precise, real-time landmark tracking without the heavy computational overhead of raw image processing (like a CNN would require). Because sign language is fundamentally a time-series problem—involving motion from a starting pose to an ending pose—an LSTM was the natural choice. LSTMs are explicitly designed to "remember" sequences over time, making them mathematically ideal for tracking the trajectory of a moving hand.

System Architecture

The "brain" of the system was an intentionally lightweight, highly customized neural network built to prevent early overfitting and handle sequential data:

- **Input Layer:** Designed to accept a matrix of (30, 225) representing 30 frames and 225 scaled 3D coordinates.
- **Masking Layer:** Added to tell the network to ignore empty or padded data.
- **LSTM Layers:** Two stacked LSTM layers, each with 256 units. The first returned the full temporal sequence, while the second condensed it into a single understanding of the movement.

- **Dense & Dropout Layers:** A fully connected Dense layer of 256 units equipped with a ReLU activation function, immediately followed by a 30% Dropout layer to forcefully randomly sever connections and force the model to generalize.
- **Output Layer:** A final Softmax Dense layer mapped to the exact number of classes (letters/signs) in the dataset, providing a percentage-based confidence score for each prediction.

The Benefits of the LSTM Pipeline

When tested in a controlled environment, this architecture showed immense promise.

- **High Theoretical Accuracy:** On a robust dataset of nearly 2,000 recorded videos, the model achieved a validation accuracy of over 91% after 150 epochs.
- **Privacy and Efficiency:** By converting raw video into abstract mathematical graphs (numbers in a .npy file), the model was highly privacy-preserving and could train rapidly on standard hardware.
- **Modularity:** The separation of the extraction math, the model blueprint, and the training loop meant the pipeline was highly organized from a software engineering standpoint.

The Challenges Faced

Developing this pipeline was a masterclass in the realities of machine learning deployment, encountering severe hurdles:

1. **Dependency Hell:** Balancing library versions across TensorFlow, Keras, and Jax to create a stable virtual environment was a significant software engineering roadblock.
2. **The "J" Bug (Static vs. Dynamic Mismatch):** Because the alphabet consists mostly of static hand shapes, the LSTM (which aggressively hunts for motion) became confused. It over-indexed on the natural, tiny tremors of the human hand, causing it to blindly guess 'J' (one of the only letters with true motion) whenever it saw an unmoving hand.
3. **The Aspect Ratio Trap:** A massive discrepancy emerged between the high-definition, 16:9 videos used to train the dataset and the standard 4:3 boxy aspect ratio of a live webcam. The math warped the coordinates, making the live hands look "stretched" to the AI.

Why We Didn't Move Forward With It

Despite excellent validation scores, the LSTM architecture was ultimately abandoned due to an insurmountable **Sim-to-Real Gap**.

The model successfully memorized the training data but catastrophically failed to generalize to unseen data (the live webcam). The LSTM had not learned the universal rules of sign language; instead, it overfitted to the specific environmental parameters of the 2,000 training videos. It learned the exact distance the signer sat from the camera, the exact speed of the movement, and the specific wobble of the hand.

When presented with unseen data—where the user sat closer to the webcam, or held a static pose rather than moving dynamically—the math completely misaligned. For a system meant to be used seamlessly in the real world, an architecture that demands the user perfectly mimic the exact lighting, scale, and speed of the training lab is not viable. The temporal strictness of the LSTM ultimately made it too rigid for the chaotic nature of live webcam inference.

Has This Been Used in PSL Before?

Yes, the MediaPipe + LSTM combination is actually a standard foundational baseline in sign language research, including Pakistan Sign Language (PSL).

However, PSL introduces an even higher level of complexity than standard one-handed alphabets. PSL relies heavily on simultaneous two-handed gestures, rapid dynamic motion, and subtle facial expressions (which require incorporating MediaPipe's face mesh). While researchers have attempted to use LSTMs for PSL, they frequently encounter the exact same generalization failures you experienced. To achieve high accuracy on unseen PSL data, modern research is increasingly pivoting away from pure LSTMs and moving toward Spatial-Temporal Graph Convolutional Networks (ST-GCNs) or Transformer models, which evaluate the spatial relationship of the joints (the shape of the hands) just as heavily as the movement over time.